

Vibration and Temperature Telemetry for Failure Prediction in Coal Handling Conveyor Belts and Mining Equipment

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DOI: 10.1016/j.coal.2025.104822 | Published: 2025

1. Abstract

Predictive maintenance in underground and opencast coal mining operations requires reliable, low-latency telemetry for continuous monitoring of critical equipment. Mechanical failures in heavy coal handling equipment, such as longwall shearers, continuous miners, and conveyor belt idler rollers, often go undetected until catastrophic failure occurs. This paper presents an integrated IoT vibration and temperature telemetry framework combined with a novel Long Short-Term Memory (LSTM) model for early failure detection.

Keywords: Predictive maintenance, Vibration telemetry, Temperature monitoring, Anomaly detection, Coal handling equipment.

2. Introduction

Mining operations rely heavily on continuous mechanical equipment throughput. Unscheduled bearing failures and thermal overload in conveyor gearboxes account for over 38% of unplanned operational halts. Traditional preventative maintenance routines miss transient micro-vibrations and localized temperature spikes.

3. Related Work

Previous studies on condition monitoring in underground mining relied on manual weekly vibration logging. Zhang et al. (2022) proposed FFT spectral analysis for fixed-speed motors, but failed under dynamic load fluctuations. Our approach extends continuous multi-sensor telemetry to dynamic mine load profiles..

4. Methodology

We deploy tri-axial MEMS accelerometers (sampling rate 10 kHz) and digital thermal RTD sensors directly on conveyor roller h
RS485 Modbus telemetry feeds real-time time-series streams into an edge computing gateway.
Feature extraction includes RMS vibration amplitude, kurtosis, crest factor, and moving-average temperature gradients..

5. Experimental Setup

Field validation was conducted over a 9-month trial at Jhanjhra Underground Mine and Rajmahal Opencast Project.

A dataset of 4.2 million time-series telemetry samples was collected across 64 operational conveyor idlers and 12 longwall power

6. Results

The proposed LSTM vibration-temperature anomaly detection model achieved an overall precision of 94.2%, recall of 91.8%, and

Early warning latency averaged 48 hours prior to catastrophic mechanical failure, reducing false alarm rate to under 2.1% across

7. Discussion & Conclusion

Integrating multi-sensor vibration and thermal telemetry significantly enhances failure prediction accuracy in harsh mining environments. The field-validated model provides reliable early warning capabilities for maintenance engineers, enabling proactive component

8. References

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